ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CCS5411 CLOUD COMPUTING

FINAL EXAMINATION

DATE: Wednesday 28th May 2026

TIME: 14:00hrs – 17:00hrs

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **TWO (2)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

CASE STUDY

Leveraging Platform as a Service (PaaS) for Digital Transformation at Central State University

Central State University (CSU) is one of Zambia's rapidly growing higher education institutions offering undergraduate and postgraduate programmes across several Schools and Faculties. Over the past decade, the University has experienced continuous growth in student enrolment, increased programme offerings, and expansion of administrative and academic operations.

To support institutional operations, the University has implemented several information systems including:

- Student Information System (SIS)
- Learning Management System (LMS)
- Financial Management System
- Human Resource Management System
- Institutional Email System
- Document Management System
- Research and Grants Management System

Although these systems have improved service delivery, they were developed and deployed independently at different times using different technologies and infrastructure environments.

As a result, CSU faces several operational challenges.

One of the major challenges is **system fragmentation**. The institutional systems operate independently and lack full interoperability. Data must frequently be entered multiple times across different systems, resulting in duplication, inconsistencies, delays in reporting, and reduced efficiency.

The University also experiences **communication challenges**. Important institutional announcements including registration deadlines, fee payment reminders, examination schedules, approval workflows, and academic notifications are often communicated manually using emails and administrative memoranda. This process creates delays and reduces responsiveness.

Furthermore, the ICT Directorate faces increasing pressure to maintain and upgrade on-premise infrastructure. Activities such as server maintenance, backup management, system availability monitoring, disaster recovery implementation, and cybersecurity management require substantial financial investment and technical expertise.

To address these challenges, the University has resolved to implement a **Platform as a Service (PaaS) cloud computing strategy**.

The proposed strategy includes:

- Hosting institutional applications in a cloud environment

- Integrating systems through APIs and middleware services
- Implementing automated communication services
- Introducing centralized identity and access management
- Improving scalability and operational efficiency

The implementation will follow these phases:

Phase I – Digital Assessment and Planning

- Infrastructure assessment
- Identification of integration requirements
- Governance and compliance review

Phase II – Cloud Platform Selection

Selection of a platform supporting:

- Application hosting
- Cloud databases
- API integration
- Notification services
- Identity management

Phase III – System Migration and Integration

- Migration of institutional services
- Centralized authentication
- Deployment of integration services

Phase IV – Communication Automation

Implementation of:

- Event-driven notifications
- Workflow automation
- Messaging services

Phase V – Governance and Capacity Building

- Staff training
- Cloud governance
- Security awareness programmes

Expected outcomes include:

- Improved institutional interoperability
- Reduced infrastructure expenditure
- Enhanced scalability and availability
- Better communication and coordination
- Improved governance and operational efficiency

SECTION A: Question 1 is compulsory and must be attempted (Practical, Scenario based or Problem based)

QUESTION 1

The University intends to implement a **Platform as a Service (PaaS)** cloud strategy to improve system integration, communication, scalability, and operational efficiency.

Provide technology recommendations and solutions for the proposed implementation.

a) Discuss ***cloud service models*** and recommend the most appropriate model for Central State University. **(6 Marks)**

b) Recommend the most ***suitable cloud service model*** for the University and justify your recommendation. **(9 Marks)**

c) Identify and explain ***FIVE (5) major components*** required in the proposed PaaS architecture. **(5 Marks)**

d) Explain how **API gateways and middleware services** support integration among institutional systems. **(5 Marks)**

e) Explain how ***centralized application hosting*** improves interoperability and service delivery. **(5 Marks)**

f) Explain how ***event-driven architecture*** supports automated institutional communication. **(4 Marks)**

g) Explain how ***messaging and notification services*** improve communication efficiency. **(3 Marks)**

h) Provide ***THREE (3) examples of institutional processes*** that can benefit from workflow automation. **(3 Marks)**

i) Discuss the following security requirements for the proposed solution:

i. ***Data encryption*** and ***privacy protection*** mechanisms. **(4 Marks)**

ii. Identity and Access ***Management controls*** suitable for institutional environments. **(3 Marks)**

iii. ***Data sovereignty***, ***audit logging*** and ***governance requirements***. **(3 Marks)**

(Total: 50 marks)

SECTION B: Attempt any TWO (2) questions in this section

QUESTION 2

Cloud storage architecture and data management are critical requirements for supporting institutional scalability and digital transformation.

a) ***Compare and contrast*** the following storage approaches:

- ***Direct Attached Storage (DAS)***
- ***Network Attached Storage (NAS)***
- ***Storage Area Network (SAN)***

Recommend the most suitable storage approach for Central State University and justify your recommendation. **(12 Marks)**

b) Explain how ***distributed cloud storage systems*** improve institutional ***performance*** and ***reliability***. **(8 Marks)**

c) Explain how ***redundancy*** and ***fault tolerance*** improve institutional business continuity. **(5 Marks)**

(Total: 25 marks)

QUESTION 3

Cloud environments require ***efficient coordination of resources*** and ***effective management of concurrent*** activities.

a) Explain the importance of **Concurrency Control** in cloud-based systems. **(5 Marks)**

b) Compare and discuss the following concurrency control protocols:

i. ***Lock-Based Protocols***
ii. ***Graph-Based Protocols***
iii. ***Timestamp-Based Protocols***

Provide examples of situations where each protocol would be appropriate. **(12 Marks)**

c) Explain how ***deadlocks*** and ***starvation*** affect cloud computing environments and recommend mitigation strategies. **(8 Marks)**

(Total: 25 marks)

QUESTION 4

Cloud adoption depends on ***virtualization technologies***, ***networking capabilities*** and ***effective infrastructure management***.

a) Explain the concept of ***virtualization*** and discuss the benefits of virtual machines in cloud environments. **(10 Marks)**

b) Discuss the following cloud characteristics and explain how each contributes to cloud service delivery:

i. ***Resource Pooling***

ii. ***Broad Network Access***

iii. ***Rapid Elasticity***

iv. ***Measured Service***

v. ***On-Demand Self-Service*** **(10 Marks)**

c) Explain the importance of ***Service Level Agreements (SLAs)*** in cloud computing environments. **(5 Marks)**

(Total: 25 marks)

END OF EXAMINATION